

AYUSH RAJ

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EXPERIENCE

Software Developer Intern ↗

Horizon Mars Rover Team, CUSAT

📅 Nov 2023–Present 📍 Kochi, Kerala

- Developed and implemented a Unified Robot Description Format (URDF) -based robotic arm model with **Arduino Due (32 bit ARM Micro-controller)** and servo motor control in Robot Operating System (ROS).
- Technologies used : **C++, ROS 2 (Humble), Linux, Arduino IDE**

AI - ML Intern ↗

IBM Skillsbuild

📅 June 2024 – July 2024 📍 Remote

- Developed expertise in neural networks, NLP, and GenAI, built a **kidney stone detector** and a **fine tuned chatbot** which was integrated into a restaurant web portal with **20% faster menu search**.
- Technologies used : **IBM Watson Studio, CSV , EJS, CSS, Bootstrap, Node.js, Express.js, MongoDB Atlas**

PROJECTS

Preventive AI ↗

📅 Sept 2024 - Oct 2024



- An AI platform combining **medical imaging, genomic data, and EHRs** for **early cancer detection** and personalized treatment insights.
- Tech Stack: **Python, Supervised ML, EJS, CSS, Node.js, Express.js, MongoDB Atlas, TensorFlow Serving**

Saarthi ↗

📅 Nov 2024



- Built a full stack application offering free educational tools, including **problem-solving resources**, an **NEP syllabus generator**, **AI chatbot**, **Doc QnA** , **Quiz**, and more.
- Tech Stack: **Bootstrap, Gemini API, Plotch AI, Node.js, Express.js, MongoDB Atlas, Vertex AI, Groclake**

Hack-Match ↗

📅 Jan 2024



- Developed a platform to help users find ideal hackathon teammates by matching them based on skill sets & interests.
- Tech Stack: **EJS, CSS, Cloudinary, Bootstrap, Render, Node.js, Express.js, NoSQL DB**

User Authentication System ↗

📅 May 2024



- Built a full-stack, responsive user authentication website with secure registration, login, and profile management
- Tech Stack: **EJS, CSS, Bootstrap, Render, Node.js, Express.js, NoSQL DB**

SOFT SKILLS

Leadership, Decision Making, Teamwork, Creativity

EDUCATION

Bachelor of Technology, Computer Science

Cochin University Of Science And Technology (CUSAT)

📍 Kochi, Kerala

📅 Oct 2022 – May 2026 (Expected)

Currently in **3rd Year** - CGPA : 9.65

Academic Achievements

- Represented **INDIA** at **European Rover Challenge, 2024, Poland** - Robotics Competition. Team name : **Horizon** (World Rank **18th**)
- Listed among **Top 10% Students** in the Batch.

POSITIONS OF RESPONSIBILITY

- Team Lead** @HORIZON- Mars Rover Team, CUSAT
- SIG AI Lead** @IEEE Execom'24- CS Branch Society,CUSAT

SKILLS

Languages

Java

Python

Javascript

C/C++

Embedded C

Frameworks and Libraries

React

OpenGL

ROS

NodeJS

ExpressJS

RestAPI

AXIOS

Gemini API

Pandas

AWS

STM32CUBE - IDE

LLMware

Databases

MongoDB

SQL

Tools

Git/GitHub

Linux

Vertex AI

OOP

AI-ML

Proteus

ACHIEVEMENTS

- Won **5 Hackathons** (out of 9) conducted by **Google Development Group** ↗ (GDG) , **Major League Hacking** ↗ (MLH) & **Inovact** ↗.
- Solved **100+** programming problems across various platforms like **LeetCode**, **Hackerrank** etc.
- Jury Choice Award Winner** at AIC by **Accenture** & **Finalist** at **DevsHouse'24 Hackathon** by **VIT Chennai**

CERTIFICATIONS

ERC Space & Robotics Competition, AGH University Poland ↗

📅 Sep 2024

Certificate of Appreciation (Tech. Team) - TEDxCUSAT↗

📅 Sep 2024

Artificial Intelligence Fundamentals, IBM ↗

📅 June 2024

Cyber Security and Ethical Hacking , IIT Delhi ↗

📅 Nov 2023

Shastra X Codechef Programming Tournament - IIT Madras ↗

📅 Oct 2023